A Digital Signature Schemes Without Using One-way Hash and Message Redundancy and Its Application on Key Agreement

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Abstract

Digital signature schemes based on public-key cryptosystems are important building blocks of key agreement protocols. They are vulnerable by existential forgery attack. A usual method to prevent existential forgery attack is on-way hash function and message redundancy. In 2004, Chang and Chang proposed a new digital signature scheme. The scheme is without using one-way hash function and any redundancy padding, which is very interesting to many designers. Recently, Chien found an existential forgery attack. In this paper, we show another forgery attack on it and propose an improved scheme without oracle, which is secure against existential forgery attacks. Then we proposed a key agreement protocol by amending the signature slightly.

1. Introduction

Digital signature schemes based on public-key cryptosystems are important building blocks of key agreement protocols. They are vulnerable by existential forgery attack. A usual method to prevent existential forgery is on-way hash function and message redundancy. Shieh et al. [1], proposed two novel multi-signature schemes: one serial multi-signature scheme and one parallel multi-signature scheme. With their high efficient of computation and communication, Sheih et al.'s multi-signature schemes are suitable for authenticating delegates in mobile codes system [2]. But, they have been shown insecure [3,4,5]. In 2004, Chang and Chang [6] proposed an improved signature scheme without using one-way hash and message redundancy, and claimed the scheme can resist forgery attacks. This is very interesting to all secure system designers. But, Chien [7] found an existential forgery attack soon. Now some researchers try to improve the scheme by modifying it slightly. Here we will also give another existential forgery attack. At the same time we point out a weakness of it. We improve the scheme. The scheme is existential forgery resistance under a weak chosen message attack.

The security of such a key agreement protocol is based on the weakest of all the cryptographic assumptions of the algorithms. If a protocol can be constructed using one cryptographic assumption, it would be at least as secure as that with multiple assumptions. In 1995, Menezes et al. proposed the first key agreement protocol that used a signature for the Diffie-Hellman public key without using a one-way hash function. In this paper, we proposed a key agreement by amending the signature slightly. For the convenience of the improved signature scheme, the key agreement protocol is constructed without using one-way hash function and message redundancy.

This paper is arranged as follows. In Section 2, we first review Chang-Chang's signature scheme, Chien's attack, some security assumptions and define our security model and some notations. Then, we propose our forgery attack on Chang-Chang's signature scheme and the improved signature scheme, at the same time we provide the security proof of it in section 3. In section 4, we present a key agreement, which is constructed by amending the signature slightly, and provide the simple security proof of it. Finally, Section 5 concludes the paper.

2. Preliminaries

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In this section, we first review Chang-Chang's signature scheme and Chien's attack. Then, we describe security assumptions and the security model.

2.1. Security notions for a signature scheme

Existential unforgeability under a weak chosen message attack require that the adversary submit all signature queries before seeing the public key. This notion is defined using the following game between a challenger and an adversary A [8]:

Query: A sends the challenger a list of q_s messages $M_1,\cdots,M_{q_s} \in \{0,1\}^*$.

Response: The challenger runs algorithm KeyGen to generate a public/ private keys pair (pk,sk). Next, the challenger generates signatures $\sigma_i = sign(sk,M_i)$ for $i=1,\cdots,q_s$.

The challenger then gives A the public key pk and the q_s signatures $\sigma_i,\cdots,\sigma_{q_s}$.

Output: Algorithm A outputs a pair (M,σ) and wins the game if

(1) M is not any of $(M_1,\cdots,M_{q_s})$, and

(2) Verify(pk,M,σ) = valid.

Definition 2.1. A forger (t,q_s,ξ)-weakly breaks a signature scheme if A runs in time at most t, A makes at most q_s signature queries, and $Adv_{w_sig_A}$ is at least ξ. A signature scheme is (t,q_s,ξ)-existentially unforgeable under a weak chosen message attack if no forger (t,q_s,ξ)- weakly breaks it. Here $Adv_{w_sig_A}$ is the probability that A wins in the above game, taken over the coin tosses of A and the challenger.

2.2. Reviews of Chang-Chang's signature scheme and Chien's attack

This section reviews Chang-Chang's digital signature scheme and Chien's attack. Let p be a large prime, and g be a primitive element in GF_p. $(x_i, Y_i \equiv g^{x_i} \bmod p)$ is the private /public key pair of U_i, here $\gcd(x_i, p-1)=1$.

Chang-Chang's signature scheme and Chien's attack can be described by table 1 and table 2 respectively.

Table 1. Chang-Chang's signature scheme

Signature
1. U computes $s \equiv Y^m \bmod p$, $k \in_R [1,p-1]$, $r \equiv m \times s \times g^{-k} \bmod p$. 2. $s+t \equiv x^{-1}(k-r) \bmod (p-1)$. 3. U sends (r,s,t) to verifier.
Verification
1. V computes $m' \equiv Y^{s+t} \times r \times g^r \times s^{-1} \bmod p$. 2. If $s \equiv Y^{m'} \bmod p$, he accepts the signature; otherwise he rejects it.

Table 2. Chien's attack

Forgery signature
1. Eve chooses $\bar{k} \in_R [1,p-1]$, computes $r \equiv Y^{\bar{k}} \bmod p$. 2. $m \equiv g^r \bmod p$, $s \equiv Y^m \bmod p$, $t \equiv m - \bar{k} - s \bmod (p-1)$. 3. Eve sends (r,s,t) to verifier.
Verification
1. V computes $m' \equiv Y^{s+t} \times r \times g^r \times s^{-1} \bmod p \equiv g^r \bmod p \equiv m$. 2. $s \equiv Y^{m'} \bmod p$ holds. The verifier accepts the forgery signature.

2.3. Security assumption

Definition 2.2. Decisional Diffie-Hellman (DDH) Problem. The decisional DH problem is to distinguish between tuples of the form $(\beta,\beta^a,\beta^b,\beta^{ab})$ and $(\beta,\beta^a,\beta^b,\beta^c)$ for random $\beta \in \mathrm{G}$, and $a,b,c \in Z_p^*$. An algorithm A is said to solve the DDH problem with an advantage of ξ if $|\Pr[A(\beta,\beta^a,\beta^b,\beta^{ab})=1] - \Pr[A(\beta,\beta^a,\beta^b,\beta^c)=1]| \le \xi$.

2.4. The security model for a key agreement protocol

The model described in this section is based on that of [9,10,11]. We modify the modes in two party without random oracle.

Participants. U_1 and U_2 are the participants. In this model we allow each user to execute a protocol many times with different users. A user may execute a polynomial number of protocol instances in parallel.

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Initialization. During this phase, which is conducted before the first execution of the key establishment protocol, the master secret key x_i and global parameters are generated by algorithm Setup. CA issue certificates for the two users. Each user gets public and private keys.

Adversarial model. Normally, the security of a protocol is related to the adversary's ability. The abilities are formally modeled by queries issued by adversaries. We assume that a probabilistic polynomial time adversary A controls the communications completely and can make queries to any instance. The list of queries that A can make is summarized below:

- Execute ($\{U_1, U_2\}$): This query executes a protocol run between the users $\{U_1, U_2\}$, and the adversary A gets the complete transcripts of all the messages sent during the protocol execution.
- Send (U_i, M): This query allows the adversary A to send a message M to instance U_i, and A gets back the reply generated by U_i upon receipt of message M, where $i = 1,2$.
- Reveal (U_i):This query models the misuse of session keys by a user. It returns the session key held by U_i.
- Corrupt (U_i): This query allows the adversary A to get the long-term private key corresponding to U_i. But the adversary A does not get the internal data of any instance of U_i executing the protocol.
- Test (U_i, t): This query tries to capture the adversary's ability to tell apart a real session key from a random one. It returns the session key if $b = 1$ or a random key of the same size if $b = 0$. We suppose the adversary can ask the query just once.

2.5. Security notations for a key agreement protocol

Let IDS_i is the session identifier of a participant instance U_i. It is a function of all the messages sent and received by U_i. The partner identifier id_i of a participant instance U_i is a set of the users with whom U_i intends to establish a session key.

Definition 2.3. Partnering. Two instance U_i and U_j are partnered if and only if $\mathrm{IDS}_i = \mathrm{IDS}_j$ and $\mathrm{id}_i = \mathrm{id}_j$.

Definition 2.4. Freshness. We say that a instance U_i is fresh if the instance goes into an *accept* state after receiving the last excepted protocol message and no *Reveal* has been asked to U_i or to its partner.

Definition 2.5. Protocol Security. We denote the advantage of the adversary A attacking the protocol as $\mathrm{Adv}_A(k) = |2 \cdot \Pr[\mathrm{Succ}]\text{-}1|$. We say the key agreement protocol secure if for all PPT adversary A $\mathrm{Adv}_A(k)$ is negligible.

Definition 2.6. Authentication. A key agreement protocol is said to provide authentication if for a user U_i, no other users except partners can learn the value of a session key.

Definition 2.7. Forward Secrecy. Forward secrecy means that an adversary gets negligible advantage in knowing information about previously established session keys when making a *Corrupt* query.

3. Our forgery attack and the improve the signature scheme

The verifier can escape Chien's forgery attack by checking if $m \equiv g^r \bmod p$. In this section we will show our forgery attack on Chang-Chang's attack. In the attack, $m \neq s, m \neq t, m \neq r$, then we improve the scheme.

3.1. Forgery attack

Signature

1. Eve chooses $w \in [0, p-1]$, $r \in GF[p]$ randomly.
2. Eve computes $k = g^r \times Y^w (\bmod p)$, and forge $m = r \times k = r \times g^r \times Y^w (\bmod p)$. Then $s = Y^m (\bmod p)$.
3. Eve derives t from $s + t = m + w (\bmod p-1)$. And Eve sends (s,t,r) to the verifier.

Verification

1. After receiving the signature from Eve, verifier computes: $Y^{s+t} \times r \times g^r \times s^{-1} = Y^m \times Y^w \times r \times g^r \times s^{-1} = m' (\bmod p)$.
2. Here $m' = m$, so $Y^{m'} = s (\bmod p)$ holds. The verifier accepts the forgery signature.

3.2. Improved signature scheme

Signature

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(1) **Setup.** U runs KeyGen, and get the public/private keys pairing (pk, sk), here $sk = x$, $pk = Y = g^x (\bmod p)$.

(2) **Sign.** U computes $s = Y^m (\bmod p)$ and $r = g^{-k} s^x (\bmod p)$, $k \in Z_p^*$ is a random number. Then he derives t from $t = (1-x)m + x^{-1}(k-r)(\bmod p-1)$.

U sends (s, r, t) to the verifier.

Verification

After receiving (s, r, t), the verifier computes $(pk)^t = s^{(1-x)} \times g^{(k-r)} (\bmod p)$.

If $(pk)^t \times r \times g^r = s$ holds, the verifier will accept the signature, otherwise he will reject it.

3.3. Security proof

Theorem 1. Suppose the (q, t', ξ) —Discrete Logarithms assumption holds in G. Then the improved signature scheme above is (q, t_s, ξ) -secure against existential forgery under a weak chosen message attack. Here $q_s < q$ and $t \le t' - o(qT)$. Where T is the maximum time for an exponentiation in G.

Proof. Assume A is a forger that (q, t_s, ξ) - breaks the scheme, then we can construct an algorithm B, by interacting with A, which can solve DH problem in G in time t' with advantage ξ. B is given a random instance (g, Y), $Y \equiv g^x \in G$, x is unknown. B can produce a pair (c, β^{cx}) for some $c \in G$, here x is unknown to B。 B interacts with A as following:

Query. A outputs distinct q_s messages, here $q_s < q$.

Response. B responses with a public key and signatures of q_s messages getting from A. Let $f(y) = \prod_{i=0}^{q_s-1}(m_i y + 1) \ne 0$ is a polynomial. We expand f and write $f(y) = \sum_{i=0}^{q_s} \alpha_i y^i \ne 0$ where $\alpha_0, \cdots, \alpha_{q_s} \in Z_p$ are the coefficients of $f(y)$. For each $i = 1, \cdots, q_s$, B generates a signature (s_i, r_i, t_i) on m_i by the following steps.

(1) B computes $g' \equiv g^{f(x)} (\bmod p)$, and the public key $Y' \equiv g^{xf(x)} (\bmod p)$. The public key given to A is (g', Y'). Let $f_i(y)$ be the polynomial $f_i(y) = (my^2 - k) f(x)$, here we expand f_i and write $f(y) = \sum_{i=0}^{q_s+2} \beta_i y^i \ne 0$.

(2) For each $i = 1, \cdots, q_s$, B produces the signature on m_i under the public key (g', Y'). Firstly, B computes $s_i' = (Y')^{m_i} \bmod p$ and $r_i' = g^{f_i(x)} = (g')^{-k_i} (s')^x (\bmod p)$. Then he computes t_i'. To do so, let $\Upsilon(y)$ be the polynomial $\Upsilon(y) = (-m_i y^2 + y + k_i - r)/y$. B $t_i' = (1-x)m_i + x^{-1}(k_i - r)(\bmod p-1) = \Upsilon(x)$. At last B sends (s_i', r_i', t_i') to A.

Output. A forge a signature on m_*, where $m_* \notin \{m_1, \cdots, m_{p-1}\}$. A sends (s_*, r_*, t_*) to B. The forgery signature can be denoted as $s_* = (Y')^{m_*}$, $r_* = g'^{(-k_*)}(s_*)^x = g^{-f(x)k_*} g^{x^2 f(x) m_*} = g^{f(x)(-k_* + m_* x^2)} \bmod p$, and $t_* = (1-x)m_* + x^{-1}(k_* - r_*)(\bmod p-1)$. B computes:

$$Y^{t_*} = Y^{(1-x)m_*} g^{k_*} g^{-r_*} \bmod p \qquad (1)$$

$$\text{and } (r_*)^{f^{-1}(x)} = g^{(-k_* + m_* x^2)} \bmod p. \qquad (2)$$

According the two equations (1) and (2), B can get equation (3).

$$Y^{t_*} g^{r_*} Y^{-m_*} = Y^{-xm_*} g^{k_*} \bmod p \qquad (3)$$

The equation (3) can be written as $g^{x(t_* - m_*) + r_*} = [(r_*)^{f^{-1}(x)}]^{-1} \bmod p$. Then B multiples g^{r_*} on the both sides of this equation. B can get $g^{x(t_* - m_*)} = [(r_*)^{f^{-1}(x)}]^{-1} g^{r_*} \bmod p$. Let $[(r_*)^{f^{-1}(x)}]^{-1} g^{r_*} = S$, B return $(t_* - m_*, S)$ as the solution of DH.

4. Key agreement protocol

In this section, we propose our two-party key agreement protocol according the improved signature scheme.

4.1. A two-party key agreement protocol

The protocol which we called T-KA, involves a CA which will issue certifications, and two entities U_i, where $i = 1, 2$.

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Setup. CA issues certifications for the two entities. Cert_i is U_i's certification. U_i's secret key is x_i, and the public key is $Y_i = g^{x_i} \bmod p$.

Interacting. U_i picks a random $a_i \in Z_p^*$ and set $r_i = Y_i^{a_i} \bmod p$. U_i computes $s_i = g^{-k_i} r_i^{x_i} (\bmod p)$, and $t_i = (1-x_i)a_i + x_i^{-1}(k_i - s_i)(\bmod p-1)$, where $k_i \in Z_p^*$ is a random number. U_i sends $\delta_i = (r_i, s_i, t_i, \mathrm{Cert}_i)$ to U_j, where $j \neq i$.

Key Computation. For $j \neq i$, upon receiving the δ_i, U_j computes $(Y_i)^{t_i} = r^{(1-x_i)} \times g^{(k_i - s_i)} (\bmod p)$. If $(Y_i)^{t_i} \times s_i \times g^{s_i} = r_i$ holds, U_j computes $K_{ji} = (r_i)^{x_j a_j} \bmod p$, otherwise he will reject it. Thus U_1 and U_2 get a common key $K = K_{12} = K_{21} = g^{x_1 x_2 a_1 a_2} \bmod p$.

4.2. Security proof

In this section, we show that the protocol is secure against an active adversary under the DDH assumption. In other words, if there exists an active adversary who has non-negligible probability of breaking the protocol, then he also has non-negligible probability of breaking the DDH assumption.

Theorem 2. The above protocol is secure against an active adversary under the DDH assumption in the standard Model. Concretely, $\mathrm{Adv}_A \le 2 \cdot q_{ex} \cdot \mathrm{Adv}^{\mathrm{DDH}}$.

Proof. Assume A is an adversary that can get an advantage in breaking T-AK. We first consider the case that the adversary makes only one Execute query and then extend this to the case that A makes multiple Execute queries. The number of users chosen by the adversary A is 1. The distribution of the transcript T and the resulting group session key K is given by:

$$\text{params}=\left[\mathrm{g} \leftarrow GF[p]; x_1, x_2 \leftarrow Z_p^*; Y_1 = g^{x_1}, Y_2 = g^{x_2}\right]$$

$$Real = \begin{bmatrix} k_1, k_2 \leftarrow Z_p^*, a_1, a_2 \leftarrow Z_p^*, \mathrm{Cert}_1, \mathrm{Cert}_1; \\ r_1 = Y_1^{a_1} \bmod p, r_2 = Y_2^{a_2} \bmod p; \\ s_i = g^{-k_i} r^{x_i} \bmod p, i = 1,2; \\ t_i = (1-x_i)a_i + x_i^{-1}(k_i - s_i)(\bmod p-1), i = 1,2; \\ T = \langle r_1, r_2, s_1, s_2, t_1, t_2, \mathrm{Cert}_1, \mathrm{Cert}_1 \rangle; \\ K = (r_1)^{x_2 a_2} = (r_2)^{x_1 a_1} = g^{x_1 x_2 a_1 a_2} \bmod p; \end{bmatrix}$$

Consider the distributions $Fake_1$ defined as follows:

$$Fake_1 = \begin{bmatrix} k_1, k_2 \leftarrow Z_p^*, a_1, a_2 \leftarrow Z_p^*, \mathrm{Cert}_1, \mathrm{Cert}_1; b_1 \leftarrow Z_p^* \\ r_1 = Y_1^{a_1} \bmod p, r_2 = Y_2^{a_2} \bmod p; \\ s_1 = g^{-k_1} r^{b_1} \bmod p, s_2 = g^{-k_2} r^{x_2} \bmod p; \\ t_1 = (1-b_1)a_1 + b_1^{-1}(k_1 - s_1)(\bmod p-1), \\ t_2 = (1-x_2)a_2 + x_2^{-1}(k_2 - s_2)(\bmod p-1); \\ T = \langle r_1, r_2, s_1, s_2, t_1, t_2, \mathrm{Cert}_1, \mathrm{Cert}_1 \rangle; \\ K = g^{b_1 x_2 a_1 a_2} \bmod p; \end{bmatrix}$$

Let $\xi = \mathrm{Adv}^{\mathrm{DDH}}$. Assume that A made q_{se} times Send queries and q_{ex} times Execute queries. Then A randomly chooses (T, K) to make a Test query and outputs b'. A can distinguish $K = g^{b_1 x_2 a_1 a_2} \bmod p$ from $K = g^{x_1 x_2 a_1 a_2} \bmod p$ with probability ξ', where $\xi' \le \xi$. Hence A can correctly guesses $b = b'$ with probability ξ'. The remaining steps continue in the same way and we obtain:

$$\xi' = |\Pr[T \leftarrow Real; K \leftarrow Real; A(T,K)] = 1 - \Pr[T \leftarrow Fake_1; K \leftarrow Fake_1; A(T,K)] = 1| \le \xi;$$

$$\mathrm{Adv}_A = |2\Pr[\mathrm{Succ}] - 1| = |2\Pr[b = b'] - 1| = 2\xi';$$

Hence, ξ' is the advantage that DDH can be solved in $\mathrm{GF}(p)$. So, we can get $\mathrm{Adv}_A \le 2q_{es}\xi'$, expanding it, we get $\mathrm{Adv}_A \le 2 \cdot q_{ex} \cdot \mathrm{Adv}^{\mathrm{DDH}}$.

5. Conclusions

Many researchers are interested in Chang- Chang's signature scheme for its convenience, and try to improve it. In this letter, we give out a new forgery attack on it and improve it. Our improved signature scheme is existentially unforgeable under a weak chosen message attack. One issue not address in the current paper is whether our protocol remains secure in the presence of an adaptive chosen message attack which we expect to address in the further version. Many system designer are very interested in Chang-Chang's digital signature scheme. So, we proposed a two party key agreement protocol by modifying the improved scheme and prove the security in standard model.

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